MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Medical Record Science I (BBA(HM) 202) — 2024-2025 Sem 2
PYQ Answers

GROUP-A — Very Short Answer Type (1 Mark Each)

i. What is the full form of MR in healthcare?

MR stands for **Medical Record**. It refers to the systematic documentation of a patient's medical history, diagnoses, treatments, and outcomes.

ii. Name any two types of medical records.

- **Inpatient Records:** Records maintained for patients admitted to the hospital.
- **Outpatient Records:** Records for patients receiving treatment without admission (OPD).

iii. Name one method used for preserving medical records.

Microfilming — a photographic process that reduces medical documents to miniature images stored on film, saving space and ensuring long-term preservation.

iv. Mention one role of an MR technician.

An MR technician is responsible for **coding diagnoses and procedures** using standardized systems like ICD (International Classification of Diseases) to ensure accurate documentation and billing.

v. What is meant by preservation of medical records?

Preservation of medical records refers to the process of maintaining and protecting medical documents from damage, decay, or loss over time using methods such as microfilming, digitization, fire-proof storage, and controlled environments.

vi. Who is responsible for managing the Medical Records Department?

The **Medical Record Officer (MRO)** or **Health Information Manager** is responsible for managing the Medical Records Department, overseeing staff, systems, and compliance.

vii. Define medical audit.

A medical audit is a systematic review and evaluation of the quality of medical care provided to patients by examining medical records against established standards to identify areas for improvement.

viii. Name any one legal document maintained in medical records.

Consent Form — a document signed by the patient or guardian giving informed permission for medical procedures or treatment.

ix. What is one characteristic of a good medical record?

Accuracy — a good medical record must contain correct, precise, and factual information about the patient's condition, diagnosis, and treatment without errors.

x. Define retention of medical records.

Retention of medical records refers to the policy of keeping medical records for a specified period of time as required by law or institutional policy before they can be destroyed or archived.

xi. What is a medico-legal case?

A medico-legal case (MLC) is a case where a patient's injury or illness has legal implications, such as accidents, assaults, poisoning, rape, or deaths under suspicious circumstances, requiring documentation for legal purposes.

xii. Name one type of record involved in medico-legal cases.

Post-Mortem Report (also called Autopsy Report) — a document prepared by a forensic pathologist detailing the cause and manner of death, used as legal evidence.

GROUP-B — Short Answer Type (5 Marks Each, Answer any 3)

Q2. Describe the different types of medical records with examples.

- **Inpatient Records:** For admitted patients — includes admission notes, progress notes, discharge summary, operative notes. *Example: Case sheets of a surgery patient.*
- **Outpatient Records:** For OPD patients — includes consultation notes, prescriptions, investigation reports. *Example: OPD card.*
- **Emergency Records:** For casualty patients — includes triage notes, emergency treatment given. *Example: Accident & Emergency register.*
- **Diagnostic Records:** Lab reports, X-rays, ECG, MRI reports. *Example: Blood test report.*

- **Medico-Legal Records:** MLCs, post-mortem reports, wound certificates. *Example: MLC register.*
- **Administrative Records:** Birth and death certificates, transfer letters. *Example: Death certificate.*

Q3. Explain the importance of medical records in legal cases.

Medical records serve as critical legal documents in various situations:

- **Evidence in court:** Records provide objective evidence of the treatment given, helping defend against negligence claims.
- **Medico-legal cases:** Essential for documenting injuries from accidents, assaults, or poisoning for police and court use.
- **Insurance claims:** Records verify the nature and extent of illness/injury for insurance processing.
- **Consent documentation:** Signed consent forms protect healthcare providers from legal liability.
- **Death investigation:** Post-mortem reports and hospital records determine cause of death in suspicious cases.
- **Patient rights:** Patients have legal right to access their own medical records.
- **Professional accountability:** Records hold medical professionals accountable for their clinical decisions.

Q4. Describe how MRD handles medico-legal documents.

The Medical Records Department handles medico-legal documents with special care:

- **Identification:** MLCs are identified and tagged at the point of entry (emergency/OPD).
- **Separate registration:** MLCs are registered in a dedicated MLC register with a unique number.
- **Secure storage:** MLC files are stored in locked, restricted-access areas.
- **Confidentiality:** Access is strictly controlled — only authorized personnel, police (with proper documentation), or courts can access MLC records.
- **Documentation completeness:** All details including injury description, time of reporting, police intimation, and treatment given must be recorded accurately.
- **Chain of custody:** Any document released is recorded with recipient details, date, and purpose.
- **Retention:** MLC records are retained for a longer period (often 10+ years) due to their legal significance.

Q5. Write a short note on assembling and deficiency checking of medical records.

Assembling: After a patient is discharged, all loose sheets (history, progress notes, lab reports, consent forms, discharge summary) are collected and arranged in a standard order as per hospital policy. This makes the record complete and easy to review.

Deficiency Checking: After assembling, each record is checked for missing or incomplete entries:

- Unsigned discharge summaries or operation notes.
- Missing diagnosis or procedure codes.
- Incomplete history and physical examination.
- Missing consent forms.
- Illegible entries.

A deficiency slip is prepared and sent to the concerned doctor or department for completion. Only complete records are filed in the permanent storage system.

Q6. Briefly describe the organizational structure of a Medical Records Department.

- **Medical Record Officer (MRO) / HIM Manager:** Head of the department, responsible for overall management, policies, and compliance.
- **Medical Record Technicians:** Handle coding (ICD), indexing, and record retrieval.
- **Medical Record Assistants:** Manage filing, assembling, deficiency checking, and clerical work.
- **Coding Staff:** Assign ICD/CPT codes to diagnoses and procedures.
- **Statistics Section:** Compile hospital statistics (bed occupancy, mortality rates, etc.).
- **Birth & Death Registration Section:** Issue and maintain birth/death certificates.
- **Medico-Legal Section:** Manage MLC records and liaison with legal authorities.

The MRD works under the Hospital Administrator and collaborates with all clinical departments.

GROUP-C — Long Answer Type (15 Marks Each, Answer any 3)

Q7. Explain various methods used in the preservation of medical records, both manual and electronic.

Preservation of medical records ensures their availability, integrity, and usability over time.

Manual Methods:

- **Microfilming:** Documents are photographed and stored on microfilm rolls. Space-saving and durable for 100+ years.
- **Lamination:** Important documents are covered with plastic film to protect from moisture and tearing.
- **Controlled storage:** Records stored in fire-proof cabinets, temperature and humidity-controlled rooms.
- **Pest control:** Regular treatment to prevent insect damage to paper records.
- **Acid-free folders:** Using archival-quality folders that don't degrade paper over time.
- **Microfiching:** Similar to microfilming but stored on flat sheets (microfiche cards).

Electronic Methods:

- **Electronic Health Records (EHR):** Digitizing all patient records into a secure electronic system.
- **Scanning and digitization:** Converting paper records to digital format (PDF/TIFF) and storing on servers.
- **Cloud storage:** Storing records on secure cloud platforms with redundancy and backup.
- **RAID systems:** Redundant Array of Independent Disks — ensures data is not lost even if one disk fails.
- **Optical discs (CD/DVD):** Archiving records on optical media for long-term storage.
- **Backup and disaster recovery:** Regular automated backups to off-site locations.
- **Encryption:** Protecting digital records with encryption to prevent unauthorized access.

Both methods must comply with legal retention periods and confidentiality requirements.

Q8(a). Elaborate on the types, process, and outcomes of medical audits. Why are they essential in hospitals?

Types of Medical Audit:

- **Prospective Audit:** Review of care before treatment is given — helps plan appropriate care.
- **Concurrent Audit:** Review while patient is still receiving care — allows immediate corrections.
- **Retrospective Audit:** Review after patient is discharged — most common type, identifies patterns over time.
- **Internal Audit:** Conducted by hospital's own staff.
- **External Audit:** Conducted by outside bodies or accreditation agencies.

Process of Medical Audit:

1. Select a topic or aspect of care to audit.
2. Set standards and criteria for good practice.
3. Collect data from medical records.
4. Compare findings against the set standards.
5. Identify gaps and areas for improvement.
6. Implement changes.
7. Re-audit to check if improvement has occurred (audit cycle).

Outcomes:

- Improved quality of patient care.
- Reduced medical errors.
- Better documentation practices.
- Enhanced accountability among clinicians.

Why Essential in Hospitals: Medical audits ensure that care meets professional standards, improve patient safety, support accreditation (NABH/JCI), and help hospitals identify and rectify clinical and administrative deficiencies systematically.

Q8(b). What are the guidelines for determining staffing levels and requirements in the MRD?

- **Bed strength:** Larger hospitals with more beds need more MRD staff.
- **Daily patient load:** OPD and IPD volumes determine workload.
- **Type of hospital:** Teaching hospitals need more staff for research and coding.
- **Services offered:** Hospitals with surgical, ICU, or specialized units generate more complex records.
- **Turnaround time standards:** If records must be retrieved within 24 hours, more staff is needed.
- **Degree of computerization:** Highly automated systems reduce manual staff requirements.
- General guideline: 1 MR technician per 50–100 beds is a commonly followed ratio.

Q9. How do MRD personnel ensure legal compliance and protect patient rights in the handling of records?

Legal Compliance:

- **Retention policy:** Records are kept for legally mandated periods (generally 5–10 years for adults, longer for minors and MLCs).

- **Authorized access only:** Records are released only to authorized persons (treating physician, patient, court, insurance with valid request).
- **MLC protocol:** Strict procedures followed for medico-legal cases, including police intimation and secure storage.
- **Coding accuracy:** Proper ICD coding ensures accurate billing and legal documentation.
- **Consent documentation:** All consent forms are filed correctly as legal evidence.

Protecting Patient Rights:

- **Confidentiality:** Patient information is not disclosed to unauthorized parties — protected under ethical and legal frameworks.
- **Right to access:** Patients have the right to obtain copies of their own records upon request.
- **Data security:** Digital records are password-protected and encrypted.
- **Correction of errors:** Patients can request correction of inaccurate information in their records.
- **Staff training:** MRD staff are regularly trained on privacy laws, ethics, and data protection.
- **Audit trails:** Electronic systems maintain logs of who accessed records and when.

Q10. How can the MRD ensure accuracy, confidentiality, and timely availability of records during digital transformation?

Ensuring Accuracy:

- Implement Electronic Health Record (EHR) systems with built-in validation checks.
- Regular deficiency checking and completion of incomplete records.
- Standardized coding using ICD-10/11 and CPT systems.
- Staff training on accurate data entry and documentation standards.
- Periodic audits to verify record completeness and accuracy.

Ensuring Confidentiality:

- Role-based access control — staff access only what their role requires.
- End-to-end encryption of patient data.
- Secure login with multi-factor authentication.
- Regular cybersecurity audits and penetration testing.
- Compliance with data protection laws.

Ensuring Timely Availability:

- Cloud-based systems allow authorized access from anywhere instantly.
- Barcode/RFID tagging of physical records for quick retrieval.
- 24/7 digital access for emergency care teams.
- Disaster recovery systems ensure continuity during system failures.

Managing Increasing Patient Volumes:

- Scalable cloud infrastructure that grows with patient load.
- Automation of routine tasks (appointment scheduling, report generation).
- Workflow management systems to distribute workload among MRD staff efficiently.

Q11. Discuss the role of medical records in healthcare delivery. Include a flow chart of their function.

Role of Medical Records in Healthcare Delivery:

- **Clinical care:** Provides physicians with complete patient history for accurate diagnosis and treatment.
- **Continuity of care:** Ensures consistent treatment when multiple providers are involved.
- **Communication tool:** Facilitates communication between doctors, nurses, and other healthcare professionals.
- **Legal protection:** Serves as evidence in medico-legal cases and malpractice suits.
- **Research and education:** Used for clinical research, teaching, and epidemiological studies.
- **Hospital administration:** Supports billing, insurance claims, resource planning.
- **Quality assurance:** Basis for medical audits and accreditation.
- **Public health:** Disease surveillance, outbreak tracking, vital statistics.

Flow Chart of Medical Record Functions:

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Patient Arrives
      │
      ▼
Registration → OPD/IPD Record Created
      │
      ▼
Clinical Consultation → History, Examination, Diagnosis Documented
      │
      ▼
Investigations → Lab/Radiology Reports Added to Record
      │
      ▼
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Treatment/Surgery → Progress Notes, Operation Notes Recorded
      │
      ▼
Discharge → Discharge Summary Prepared
      │
      ▼
MRD Assembling → All Documents Collected & Arranged
      │
      ▼
Deficiency Checking → Incomplete Entries Flagged & Completed
      │
      ▼
Coding (ICD) → Diagnoses & Procedures Coded
      │
      ▼
Indexing & Filing → Record Stored in MRD
      │
      ▼
Retrieval (when needed) → Clinical Care / Legal / Research / Audit
```